

编译原理第十二次作业参考答案

练习 8.1.1

指令	类型	代价
LD R1, #0	其他指令 (立即数载入)	1
LD R2, #0	其他指令 (立即数载入)	1
LD R3, n	访存指令	3
SUB R4, R2, R3	其他指令 (寄存器操作)	1
BGEZ R4, L2	分支指令	2
ADD R1, R1, R2	其他指令 (寄存器操作)	1
ADD R2, R2, #2	其他指令 (寄存器操作)	1
BR L1	分支指令	2
总代价		12

其他答案合理即可，比如把立即数载入指令算作访存指令，得到总代价 16

练习 8.1.2

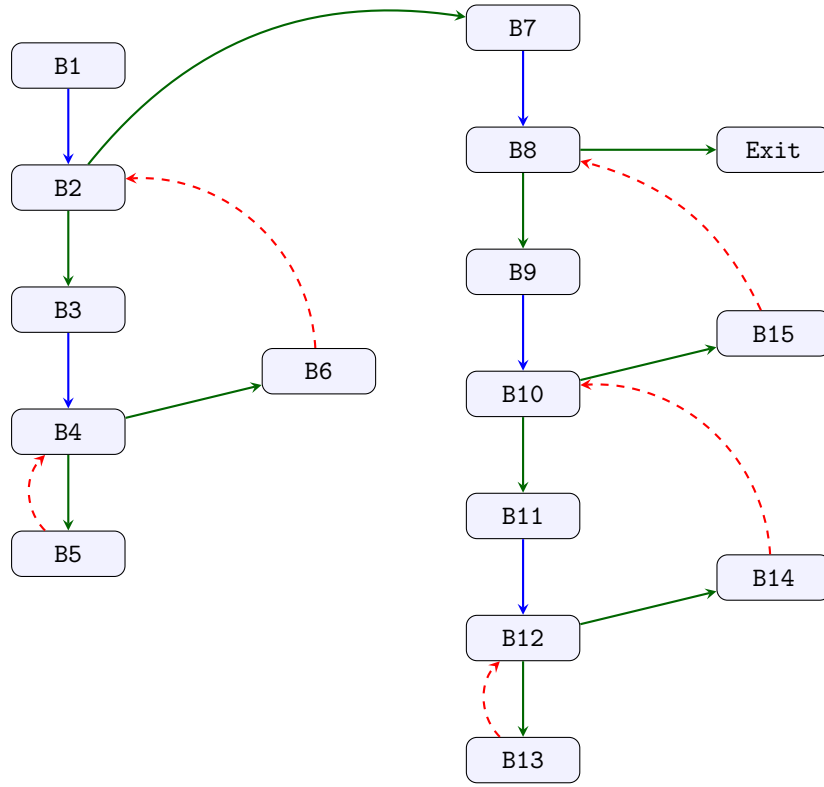
目标汇编指令	说明
LD R1, offset_i(SP)	载入 i
MUL R1, R1, #4	$i \times 4$
ADD R1, R1, SP	$R1 = SP + i \times 4$
LD R2, offset_a(R1)	$R2 = a[i]$ (此即变量 x 的值)
ST offset_x(SP), R2	将 x 写入栈帧
LD R3, offset_j(SP)	载入 j
MUL R3, R3, #4	$j \times 4$
ADD R3, R3, SP	$R3 = SP + j \times 4$
LD R4, offset_b(R3)	$R4 = b[j]$ (此即变量 y 的值)
ST offset_y(SP), R4	将 y 写入栈帧
ST offset_a(R1), R4	$a[i] = y$
ST offset_b(R3), R2	$b[j] = x$

练习 8.1.3

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基本块	三地址语句序列
B1	(1) $i = 0$
B2	(2) $\text{if } i \geq n \text{ goto } (13)$
B3	(3) $j = 0$
B4	(4) $\text{if } j \geq n \text{ goto } (11)$
B5	(5) $t1 = n * i$ (6) $t2 = t1 + j$ (7) $t3 = t2 * 4$ (8) $c[t3] = 0.0$ (9) $j = j + 1$ (10) $\text{goto } (4)$
B6	(11) $i = i + 1$ (12) $\text{goto } (2)$
B7	(13) $i = 0$
B8	(14) $\text{if } i \geq n \text{ goto } (40)$
B9	(15) $k = 0$
B10	(16) $\text{if } k \geq n \text{ goto } (38)$
B11	(17) $j = 0$
B12	(18) $\text{if } j \geq n \text{ goto } (36)$
B13	(19) $t4 = n * i$ (20) $t5 = t4 + j$ (21) $t6 = t5 * 4$ (22) $t7 = c[t6]$ (23) $t8 = n * i$ (24) $t9 = t8 + k$ (25) $t10 = t9 * 4$ (26) $t11 = a[t10]$ (27) $t12 = n * k$ (28) $t13 = t12 + j$ (29) $t14 = t13 * 4$ (30) $t15 = b[t14]$ (31) $t16 = t11 * t15$ (32) $t17 = t7 + t16$ (33) $c[t6] = t17$ (34) $j = j + 1$ (35) $\text{goto } (18)$
B14	(36) $k = k + 1$ (37) $\text{goto } (16)$
B15	(38) $i = i + 1$ (39) $\text{goto } (14)$

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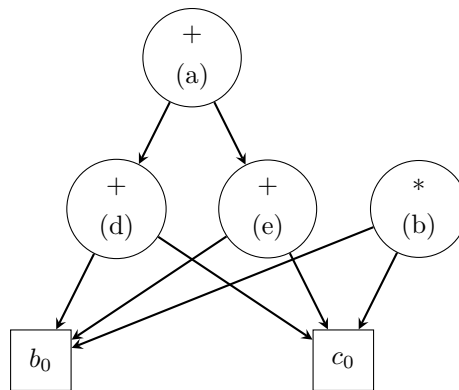


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- $\{B_4, B_5\}$
- $\{B_2, B_3, B_4, B_5, B_6\}$
- $\{B_{12}, B_{13}\}$
- $\{B_{10}, B_{11}, B_{12}, B_{13}, B_{14}\}$
- $\{B_8, B_9, B_{10}, B_{11}, B_{12}, B_{13}, B_{14}, B_{15}\}$

练习 8.1.4

1



2

$d = b + c$

$a = d + d$

公共子表达式消除，死代码消除